Throttle Advance/Limit (page 22)

Body ⇨ Truck Message #2/3, GPM 23*

(Messages available as of control unit version production month 10.2011)

Identifier: 0CFE60C9 hex
Transmission repetition time: 100 ms ± 10 ms
Data length: 8 bytes

Byte 1, 2 Requested engine speed
3, 4 Requested engine speed upper limit
5, 6 Requested engine speed lower limit
7 Requested engine torque limit
8 Requested vehicle speed limit

Description of the Parameters

Function / Type	Description of the request	Purpose/application	Resolution/Range
Requested engine speed Solldrehzahlvorgabe (PSM) Status	This is the engine speed which the engine is expected to operate at.	This information allows body applications to request a certain engine speed, e.g. from the rear of a truck mixer when a PTO is in use.	2 bytes 0.125 rpm/bit 0 rpm offset 0-8031.875 rpm
Requested engine speed upper limit Drehzahlbegrenzung max. (PSM) Status	The maximum engine speed which the engine is requested not to exceed.	This data allows body applications to request an upper engine speed limit, e.g. when a PTO is engeaged.	2 bytes 0.125 rpm/bit 0 rpm offset 0-8031.875 rpm
Requested engine speed lower limit Drehzahlbegrenzung min. (PSM) Status	The minimum engine speed which the engine is requested to allow.	This data allows body applications to request a lower engine speed limit, e.g. when a PTO is engaged.	2 bytes 0.125 rpm/bit 0 rpm offset 0-8031.875 rpm

Engine RPM, Percent Torque (page 6/7)

Truck ⇨ Body Message #1/3, GPM 13*

(Messages available as of control unit version production month 10.2011)

Identifier: 0CFE5FEB hex
Transmission repetition time: 50 ms ± 5 ms
Data length: 8 bytes

Byte 1	Towing vehicle general function	Bit	1 - 4 Engine torque mode 5 - 6 Engine control allowed 7 - 8 Engine running
2	Driver's demand engine percent torque		
3	Actual engine percent torque		
4, 5	Engine speed		
6	Percent load at current speed		
7, 8	Vehicle speed		

Description of the Parameters

Function / Type	Description of the request	Purpose/application	Resolution/Range
Driver's demand engine percent torque (CPC) Measured	The requested torque output of the engine by the driver. The date is transmitted in indicated torque as a percent of the indicated peak engine torque.	This will be necessary when making the body application's behaviour depended on the demanded engine torque, e.g. controlling of an aggregate.	1 byte 1 % /bit 125 % offset -125-125 %
Engine speed (CPC) Measured	Actual engine speed.	Informs body systems about the actual engine speed, e.g. this is necessary, if a body application should only work at a certain engine speed.	2 bytes 0.125 rpm/bit 0 rpm offset 0-8031.875 rpm

Current Gear (page 8)

Truck ⇨ Body Message #1/4, GPM 14*

(Botschaften verfügbar ab Steuergeräteversion Produktionsmonat 10.2011)

Identifier:	18FE61EB hex
Transmission repetition time:	100 ms ± 10 ms
Data length:	8 bytes

Byte		Bit
1	Percent clutch slip	
2	Current gear	
3	Towing vehicle general function	1 - 2 First clutch dependend PTO feedback 3 - 4 Second clutch dependend PTO feedback 5 - 6 Clutch independend PTO feedback 7 - 8 First engine mounted PTO feedback
4	Towing vehicle general function	1 - 2 Second engine mounted PTO feedback 3 - 4 PTO control allowed 5 - 7 Torque converter oil temperature warning 8 Not yet defined
5, 6	Torque converter oil temperature	
7	Towing vehicle general function	1 - 2 Starter active 3 - 4 Accelerator pedal low idle switch 5 - 8 Not yet defined
8	Accelerator pedal position	

Description of the Parameters

Function / Type	Description of the request	Purpose/application	Resolution/Range
Current gear ... -2 = 2. reverse gear -1 = 1. reverse gear 0 = neutral 1 = crawler 2 = 1. forward gear 3 = 2. forward gear ... (CPC) Measured	The gear currently engaged in the transmission or the last gear engaged while the transmission is in process of shifting to the new or selected gear. Transitions toward a destination gear will not be indicated. Once the selected gear has been engaged the current gear will reflect that gear.	Informs body systems which gear is currently engaged, e.g. certain truck/vehicle applications are only allowed to be in use, when the current gear is neutral.	1 byte 1 gear value/bit 125 offset -125-125 Negative values are reverse gears, positive gears are forward gears, zero is neutral.

Allison Trans Temp (page 8/9)

Truck ⇨ Body Message #1/4, GPM 14*

(Botschaften verfügbar ab Steuergeräteversion Produktionsmonat 10.2011)

Identifier:	18FE61EB hex
Transmission repetition time:	100 ms ± 10 ms
Data length:	8 bytes

Byte	1	Percent clutch slip		
	2	Current gear		
	3	Towing vehicle general function	Bit	1 - 2 First clutch dependend PTO feedback
				3 - 4 Second clutch dependend PTO feedback
				5 - 6 Clutch independend PTO feedback
				7 - 8 First engine mounted PTO feedback
	4	Towing vehicle general function	Bit	1 - 2 Second engine mounted PTO feedback
				3 - 4 PTO control allowed
				5 - 7 Torque converter oil temperature warning
				8 Not yet defined
	5, 6	Torque converter oil temperature		
	7	Towing vehicle general function	Bit	1 - 2 Starter active
				3 - 4 Accelerator pedal low idle switch
				5 - 8 Not yet defined
	8	Accelerator pedal position		

Description of the Parameters

Function Type	Description of the request	Purpose/application	Resolution/Range
Torque converter oil temperature warning (SCA) Measured	Signal which indicates that the torque converter oil temperature has reached its warning level.	This data will be necessary to make the body application's behaviour dependend on the truck's torque converter oil temperature. E.g. the application turns off, if the torque converter oil temperature has reached its warning level.	000 = No warning 001 = Prewarning 010 = Warning 011...110 = Not yet defined 111 =Not available
Torque converter oil temperature (SCA) Measured	Temperature of the torque converter lubricant.	This data will be necessary to make the body application's behaviour dependend on the truck's torque converter oil temperature. E.g. the application turns off, if the torque converter oil temperature has reached a critical value.	2 bytes 0.03125 °C /bit 273 °C offset -273-1735 °C

Allison PTO Enable/Consent

PTO Control could be realized through PSM, but as the PTO is installed and controlled by Heil, I do not understand the background – has to be discussed further.

FMVSS Lighting (page 32 to 34)

Body ⇨ Truck Message: GPM2S

Possible requests of body manufacturer

Identifier:	14EF01C9 hex
Transmission repetition time:	100 ms ±10 ms
Data length:	8 bytes

Byte	1-4	Not yet defined				
Byte	5		Bit	1–2	BB2_1_Warnblinker_Rq	Hazard warning flasher
			Bit	3–4	BB2_2_Hupe_Rq	Horn
			Bit	5–6	BB2_3_Bremslicht_Rq	Brake lights
			Bit	7–8	BB2_4_Kl58_Rq	Tml. 58
Byte	6		Bit	1–2	BB2_5_Nebelschlussleuchte_Rq	Rear fog lamp
			Bit	3–4	BB2_6_KL58und56_Rq	Tml. 58 + 56
			Bit	5–6	BB2_7_SmZusatzLeuchtausfall_Rq	Combined notification of light source failure of additional exterior lighting
			Bit	7–8	BB2_8_SmZusatzBlinkerausfall_Rq	Combined notification of light source failure of additional turn signals
Byte	7		Bit	1–2	BB2_9_FaltuerOeffnen_Rq	Folding door operation (open folding door)
			Bit	3–4	BB2_10_RKL_Rq	Rotating beacon
			Bit	5–6	BB2_11_FalttuerSchliessen_Rq	Folding door operation (close folding door)
Byte	8	Not yet defined				

Description of the Parameters

Signal name	Assignment[1]	Value	Binary	Short	Meaning	Remarks	Description[2]
BB2_1_Wamblinker_Rq	Hazard waming flasher					No body manufacturer request Hazard waming flasher request	The body manufacturer can switch on the hazard warning light with this signal. If a request is received either from the body manufacturer or from the driver, the hazard warning lights are active. The hazard warning flasher is only inactive if no requests are received from either. If the hazard warning flasher is already activated through other means, it cannot be switched off by the body manufacturer. The hazard warning light is deactivated if EMERGENCY OFF or the stealth lights are switched on. RQ is ignored. **

(more on pages 33 and 34)

Vehicle Speed (page 6/7)

Truck ⇨ Body Message #1/3, GPM 13*

(Messages available as of control unit version production month 10.2011)

Identifier:	0CFE5FEB hex
Transmission repetition time:	50 ms ± 5 ms
Data length:	8 bytes

Byte 1	Towing vehicle general function	Bit	1 - 4 Engine torque mode 5 - 6 Engine control allowed 7 - 8 Engine running
2	Driver's demand engine percent torque		
3	Actual engine percent torque		
4, 5	Engine speed		
6	Percent load at current speed		
7, 8	Vehicle speed		

Description of the Parameters

Function Type	Description of the request	Purpose/application	Resolution/Range
Vehicle speed (CPC) Measured	Speed of vehicle as calculated from tailshift speed or taken from tachograph.	Informs body systems about the vehicle speed, e.g. this is necessary, if a body application should only work below a certain vehicle speed.	2 bytes 1/256 km/h/bit; 0 km/h offset 0-251 km/h

Park brake / service brake (page 13)

Truck ⇨ Body Message , GPM 1F

(Messages available as of control unit version production month 10.2013)

Identifier:	14EF80EB hex
Transmission repetition time:	100 ms ± 10 ms
Data length:	8 bytes

Byte	1	Adaptive actuation dynamics	Bit 1-2	Parking Brake
		Brake and Chassis Functions	Bit 3-4	Service Brake
		Powertrain Functions	Bit 5-6	Clutch Current State
		Brake and Chassis Functions	Bit 7-8	Accelerator Pedal Kickdown
	2	Transmission Selected Gear		
	3	Engine Percent Torque		
	4	Nominal Engine Friction Percent Torque		
	5	Actual Maximum Available Engine Percent Torque		
	7, 8	Battery Voltage		

Description of the Parameters

Function **Type**	**Description of the request**	**Purpose/application**	**Resolution/Range**
Parking Brake (CPC) Measured	Parking Brake Status	Signal indicates the Park Brake State.	00 = Not active (No activation) 01 = Active (Activate) 10 = not defined 11 = Signal not available
Service Brake (CPC) Measured	Service Brake Status	Indicates if the Service Brake switch pedal is released or pressed.	00 = Not pressed 01 = Pressed 10 = Error 11 = Signal not available

Ambient Temperature (page 12)

Truck ⇨ Body Message #1/6, GPM 16*

(Messages available as of control unit version production month 10.2011)

Identifier: 18FE65EB hex
Transmission repetition time: 1000 ms ± 100 ms
Data length: 8 bytes

Byte 1, 2 Ambient air temperature
3- 8 Not yet defined

Description of the Parameters

Function **Type**	**Description of the request**	**Purpose/application**	**Resolution/Range**
Ambient air temperature -50 °C bis 75 °C (CPC) Measured	Temperature of air surrounding vehicle.	This data informs body systems about the temperature of air surrounding vehicle.	2 byte 0,03125 °C /bit 273 °C offset -273 – 1735 °C

Active Shift Control Indicator

The EconicSD is only available with one (primary) shift console.